SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA 0606 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15 **Total Marks:25**

Annexure A

Analytical Problem Solving

Code of Conduct:

I S.ROHINI / 192424413 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.Rohini

Annexure A

Short Answer Test

1. A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is –0.5. For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails. (5 Marks)

Concept

A Perceptron is the simplest type of artificial neural network used for binary classification problems. It takes multiple input values, multiplies them by their respective weights, adds a bias value, and applies a step activation function to classify the output as either 0 or 1.

Given Data:

- Weight for Study Hours (w_1) = 0.6
- Weight for Attendance (w_2) = 0.4
- Bias (b) = –0.5
- Study Hours (x_1) = 2
- Attendance (x_2) = 1

Formula

Net Input

$$\boldsymbol{Net = w_1 x_1 + w_2 x_2 + b}$$

Step Activation Function

- If Net ≥ 0, Output = 1
- If Net < 0, Output = 0

Calculation

$$Net = (0.6 \times 2) + (0.4 \times 1) - 0.5$$
$$= 1.2 + 0.4 - 0.5$$
$$= 1.1$$

Since

$$1.1 \geq 0$$

Output = 1

Result

The perceptron produces an output of 1.

Conclusion:

Since the output is 1, the perceptron classifies the student as Pass. This indicates that the weighted contribution of study hours and attendance is sufficient to cross the decision boundary.

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2. A deep learning model is developed for disease prediction and its performance is evaluated using a confusion matrix. The model yields 180 true positives, 150 true negatives, 20 false positives, and 50 false negatives. Using these values, compute the accuracy, precision, sensitivity (recall), and specificity of the model. Based on the calculated results, discuss whether the model is reliable for medical diagnosis, considering the importance of minimizing diagnostic errors. (5 Marks)

Concept

A Confusion Matrix is used to evaluate the performance of a classification model. It compares the predicted results with the actual results and helps calculate important evaluation metrics such as Accuracy, Precision, Recall (Sensitivity), Specificity, and F1-Score.

Given Data:

- True Positives (TP) = 180
- True Negatives (TN) = 150
- False Positives (FP) = 20
- False Negatives (FN) = 50

Total Samples

$$Total = TP + TN + FP + FN$$
$$= 180 + 150 + 20 + 50$$
$$= 400$$

1. Accuracy

Formula

$$\boldsymbol{Accuracy = \frac{TP + TN}{Total}}$$

Calculation

$$= \frac{180 + 150}{400}$$
$$= \frac{330}{400}$$
$$= 0.825$$

Accuracy = 82.5%

2. Precision

Formula

$$\boldsymbol{Precision = \frac{TP}{TP + FP}}$$

Calculation

$$= \frac{180}{180 + 20}$$

$$= \frac{180}{200}$$

$$= 0.90$$

Precision = 90%

3. Recall (Sensitivity)

Formula

$$\boldsymbol{Recall = \frac{TP}{TP + FN}}$$

Calculation

$$= \frac{180}{180 + 50}$$

$$= \frac{180}{230}$$

$$= 0.7826$$

Recall = 78.26%

4. Specificity

Formula

$$\boldsymbol{Specificity = \frac{TN}{TN + FP}}$$

Calculation

$$= \frac{150}{150 + 20}$$

$$= \frac{150}{170}$$

$$= 0.8824$$

Specificity = 88.24%

5. F1-Score

Formula

$$\boldsymbol{F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}}$$

Calculation

$$= \frac{2 \times 0.90 \times 0.7826}{0.90 + 0.7826}$$

$$= 0.837$$

F1-Score = 83.7%

Conclusion:

The model achieves 82.5% accuracy, 90% precision, 78.26% recall, 88.24% specificity, and 83.7% F1-score. Although the model performs well overall, its recall is comparatively lower, indicating that some positive cases are missed. For medical diagnosis, improving recall is important to reduce false negatives.

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3. In a neural network, a single neuron uses the delta learning rule. The initial weight is 0.4, learning rate is 0.2, and input is 3. The actual output is 0.5, while the target output is 0.9. Calculate the error, determine the weight update, and compute the new weight after one iteration. (5 Marks)

Concept

The Delta Learning Rule is a supervised learning algorithm used to update the weights of a neuron by minimizing the error between the target output and the actual output. The weight adjustment depends on the learning rate, input value, and error.

Given Data:

- Initial Weight = 0.4
- Learning Rate = 0.2
- Input = 3
- Actual Output = 0.9
- Target Output = 0.5

Formula

Error

$$\boldsymbol{Error = Target - Actual}$$

Weight Update

$$\boldsymbol{\Delta w = \eta \times Error \times Input}$$

New Weight

$$\boldsymbol{w_{new} = w + \Delta w}$$

Calculation

Step 1: Calculate Error

$$Error = 0.5 - 0.9 = -0.4$$

Step 2: Calculate Weight Update

$$\Delta w = 0.2 \times (-0.4) \times 3$$
$$= -0.24$$

Step 3: Calculate New Weight

$$w_{new} = 0.4 + (-0.24)$$
$$= 0.16$$

Result

- Error = −0.4
- Weight Change = −0.24
- Updated Weight = 0.16

Conclusion:

The weight decreases because the actual output is greater than the target output. This adjustment helps reduce prediction error in the next training iteration and improves the learning performance of the neuron.

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4. A Genetic Algorithm is used to maximize the function f(x) = 2x + 3 for an initial population of values {1, 3, 5, 7}. Compute the fitness of each individual and select the top two candidates. Perform a single-point crossover to generate offspring and explain how mutation can help avoid premature convergence. (5 Marks)

Concept

A Genetic Algorithm (GA) is an optimization technique inspired by natural selection. It searches for the best solution by repeatedly applying selection, crossover, and mutation operations on a population of candidate solutions.

Given Data:

Fitness Function

$$f(x) = 2x + 3$$

Population

$$\{1, 3, 5, 7\}$$

Step 1: Calculate Fitness

Individual	Fitness Calculation	Fitness
1	2(1)+3	5
3	2(3)+3	9
5	2(5)+3	13
7	2(7)+3	17

Step 2: Selection

The individuals with the highest fitness values are selected.

Selected Parents:

- 7 (Fitness = 17)
- 5 (Fitness = 13)

Step 3: Single-Point Crossover

Binary Representation

7 = 111

5 = 101

After crossover,

Parent 1 = 111

Parent 2 = 101

Offspring 1 = 110 (6)

Offspring 2 = 101 (5)

Step 4: Mutation

Mutate one bit of 110

110 → 111

Mutated Offspring = 7

Conclusion:

The Genetic Algorithm successfully selects the fittest individuals and generates better offspring using crossover and mutation. Mutation helps maintain population diversity and prevents premature convergence to local optimum solutions.

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5. During training of a neural network, the training loss decreases continuously, while validation loss decreases initially and then starts increasing after several epochs. Identify the issue occurring in the model. Explain the reason behind this behavior and suggest two techniques such as regularization or early stopping to improve generalization performance. (5 Marks)

Concept:

During neural network training, both training loss and validation loss are monitored. A good model should have decreasing training and validation losses. If training loss continues to decrease while validation loss starts increasing, the model is overfitting.

Observation:

- Training loss decreases continuously.
- Validation loss decreases initially.
- Validation loss increases after several epochs.

Issue Identified:

The model is suffering from Overfitting.

Reason:

The neural network learns not only the important patterns in the training data but also the noise and unnecessary details. As a result, it performs well on training data but poorly on unseen validation data, reducing its generalization ability.

Techniques to Improve Performance

1. Regularization (L1/L2):

Regularization adds a penalty term to the loss function, reducing the complexity of the model and preventing it from fitting noise.

2. Early Stopping:

Training is stopped when the validation loss starts increasing. This prevents the model from over-training and improves generalization.

Importance of Validation Accuracy:

Validation accuracy measures how well the model performs on unseen data. It is a reliable indicator of the model's ability to generalize and helps determine the optimal number of training epochs.

Conclusion:

The model exhibits overfitting because training performance improves while validation performance worsens. Applying regularization and early stopping can reduce overfitting and improve the model's ability to make accurate predictions on new data.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**25**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand